

Qualification Pack



Aerospace CNC Programmer

QP Code: AAS/Q1002

Version: 1.0

NSQF Level: 4

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AAS/Q1002: Aerospace CNC Programmer

Brief Job Description

Aerospace CNC Programmer is responsible for generating program as per CNC programming language by interpreting CAD/ CAM drawings/ models in specified file format and carrying out of verification of tool path

Personal Attributes

The individual should have basic communication ability and should be responsible in order to maintain procedures. He should possess numerical and computational abilities, openness to learning, ability to plan and organise own work and identify and solve problems in the course of working, understanding the need to take initiative and manage self and work to improve efficiency and effectiveness

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [AAS/N0503: Work Effectively in a Team](#)
2. [AAS/N1001: Follow organisation safety and security procedures](#)
3. [AAS/N1005: Carry-out pre-programming activities for aircraft parts](#)
4. [AAS/N1006: Create CNC programs for aircraft parts](#)
5. [AAS/N1007: Perform post-programming activities for aircraft parts](#)
6. [ASC/N0021: Maintain 5S at the work premises](#)
7. [ASC/N9801: Organize work and resources \(Service\)](#)
8. [ASC/N9802: Interact effectively with colleagues, customers and others](#)

Qualification Pack (QP) Parameters

Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Machining
Country	India

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NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2144.0900
Minimum Educational Qualification & Experience	Diploma (Mechanical Engineering)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	15/05/2018
Next Review Date	03/11/2027
NSQC Approval Date	03/11/2022
Version	1.0
Reference code on NQR	2019/AA/AASSC/02967
NQR Version	1

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AAS/N0503: Work Effectively in a Team

Description

This OS describes the skills and knowledge required to work effectively within and with teams.

Scope

The scope covers the following :

- Support the work team

Elements and Performance Criteria

Support the work team

To be competent, the user/individual on the job must be able to:

- PC1.** display courteous and helpful behaviour at all times.
- PC2.** take opportunities to enhance the level of assistance offered to colleagues.
- PC3.** meet all reasonable requests for assistance within acceptable workplace timeframes.
- PC4.** complete allocated tasks as required.
- PC5.** seek assistance when difficulties arise.
- PC6.** use questioning techniques to clarify instructions or responsibilities.
- PC7.** identify and display a non discriminatory attitude in all contacts with customers and other staff members.
- PC8.** observe appropriate dress code and presentation as required by the workplace, job role and level of customer contact.
- PC9.** follow personal hygiene procedures according to organisational policy and relevant legislation
- PC10.** interpret, confirm and act on workplace information, instructions and procedures relevant to the particular task.
- PC11.** interpret, confirm and act on legal requirements in regard to anti- discrimination, sexual harassment and bullying.
- PC12.** ask questions to seek and clarify workplace information.
- PC13.** plan and organise daily work routine within the scope of the job role.
- PC14.** prioritise and complete tasks according to required timeframes.
- PC15.** identify work and personal priorities and achieve a balance between competing priorities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** ask questions to identify and confirm requirements.
- KU2.** follow routine instructions through clear and direct communication
- KU3.** use language and concepts appropriate to cultural differences

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- KU4.** use and interpret non-verbal communication.
- KU5.** the scope of information or materials required within the parameters of the job role.
- KU6.** consequences of poor team participation on job outcomes.
- KU7.** work health and safety requirements.
- KU8.** The policies and procedures relating to the job role.
- KU9.** The value system of the organisation.
- KU10.** Employee rights and obligations.
- KU11.** The reporting hierarchy and escalation matrix.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation accurately
- GS2.** write simple reports when required.
- GS3.** read information accurately.
- GS4.** read and interpret data sheets.
- GS5.** listen to and orally communicate information with all concerned
- GS6.** make appropriate decisions regarding the responsibilities of the job role.
- GS7.** monitor efficient functioning of all activities
- GS8.** plan and organise work to achieve targets and deadlines
- GS9.** communicate with passengers and other stakeholders in a courteous manner
- GS10.** maintain effective work relationship
- GS11.** identify trends/common causes for errors and suggest possible solutions to the supervisor / management
- GS12.** identify and correct errors
- GS13.** analyse best possible solutions (cost, time, effort, etc.) suited for operations
- GS14.** concentrate on task at hand and complete it without errors
- GS15.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Support the work team</i>	45	55	-	-
PC1. display courteous and helpful behaviour at all times.	3	3	-	-
PC2. take opportunities to enhance the level of assistance offered to colleagues.	3	4	-	-
PC3. meet all reasonable requests for assistance within acceptable workplace timeframes.	3	3	-	-
PC4. complete allocated tasks as required.	3	3	-	-
PC5. seek assistance when difficulties arise.	3	4	-	-
PC6. use questioning techniques to clarify instructions or responsibilities.	3	3	-	-
PC7. identify and display a non discriminatory attitude in all contacts with customers and other staff members.	3	3	-	-
PC8. observe appropriate dress code and presentation as required by the workplace, job role and level of customer contact.	3	4	-	-
PC9. follow personal hygiene procedures according to organisational policy and relevant legislation	3	4	-	-
PC10. interpret, confirm and act on workplace information, instructions and procedures relevant to the particular task.	3	4	-	-
PC11. interpret, confirm and act on legal requirements in regard to anti- discrimination, sexual harassment and bullying.	3	4	-	-
PC12. ask questions to seek and clarify workplace information.	3	4	-	-
PC13. plan and organise daily work routine within the scope of the job role.	3	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. prioritise and complete tasks according to required timeframes.	3	4	-	-
PC15. identify work and personal priorities and achieve a balance between competing priorities	3	4	-	-
NOS Total	45	55	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N0503
NOS Name	Work Effectively in a Team
Sector	Aerospace and Aviation
Sub-Sector	Airline
Occupation	Loading & Unloading
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	20/02/2017
Next Review Date	17/06/2023
NSQC Clearance Date	17/12/2022

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AAS/N1001: Follow organisation safety and security procedures

Description

This unit is about maintaining awareness of safety and security procedures as defined by organisational policies and regulatory guidelines

Elements and Performance Criteria

Element Performance Criteria Comprehending the safety and security procedures for conduct of operations, data security, approvals, health, safety and security

To be competent, the user/individual on the job must be able to:

- PC1.** comply with the organisations safety and security policies and procedures
- PC2.** comply with the regulatory guidelines on safe conduct of operations and maintenance of conditions to thwart any acts of unlawful interference
- PC3.** report any identified breaches of safety and security policies and procedures to the designated person
- PC4.** report any theft of organisation property according to the organisation policy
- PC5.** coordinate with other resources at the workplace (within and outside the organisation) to achieve a safe and secure environment
- PC6.** identify and mitigate any safety and security hazards like illness, accidents, fires or acts of unlawful interference if it falls within the limits of the individuals authority
- PC7.** report any hazards outside the individuals authority to the relevant person in line with organisational procedures and regulatory guidelines
- PC8.** follow the organisations emergency procedures for accidents, fires or acts of unlawful interference
- PC9.** identify and recommend opportunities for improving health, safety, and security to the designated person
- PC10.** ensure that all health and safety records are updated and procedures are well defined

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** hazard identification and risk management as defined within the organisational policy and procedures
- KU2.** legislative requirements and organisations procedures for maintenance of safety and security standards and individuals role and responsibilities in relation to this
- KU3.** how and when to report hazards
- KU4.** the limits of responsibility for dealing with hazards
- KU5.** the organisations emergency procedures for different emergency situations and the importance of following these
- KU6.** the importance of maintaining high standards of safety and security
- KU7.** implications that any non-compliance with safety and security may have on individuals and the organisation

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- KU8.** different types of breaches of safety and security and how and when to report these
- KU9.** evacuation procedures for workers
- KU10.** how to summon medical assistance and the emergency services, where necessary
- KU11.** how to use the health, safety and accident reporting procedures and the importance of these
- KU12.** regulatory guidelines on dealing with safety and security emergencies

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete accurate, well written report in English language detailing the situations of emergency with attention to detail
- GS2.** read instructions, guidelines/procedures/rules
- GS3.** listen to and orally communicate information with all concerned
- GS4.** make decisions on a suitable course of action or response if permitted by the authority matrix
- GS5.** monitor efficient functioning of all activities
- GS6.** plan and organise work to achieve targets and deadlines
- GS7.** communicate with customers in a courteous manner
- GS8.** maintain effective relationship with the customers
- GS9.** identify trends/common causes for errors and suggest possible solutions to the supervisor / management
- GS10.** identify and correct errors
- GS11.** analyse best possible solutions (cost, time, effort, etc) suited for operations
- GS12.** concentrate on task at hand and complete it without errors
- GS13.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Element Performance Criteria Comprehending the safety and security procedures for conduct of operations, data security, approvals, health, safety and security</i>	40	60	-	-
PC1. comply with the organisations safety and security policies and procedures	4	6	-	-
PC2. comply with the regulatory guidelines on safe conduct of operations and maintenance of conditions to thwart any acts of unlawful interference	4	6	-	-
PC3. report any identified breaches of safety and security policies and procedures to the designated person	4	6	-	-
PC4. report any theft of organisation property according to the organisation policy	4	6	-	-
PC5. coordinate with other resources at the workplace (within and outside the organisation) to achieve a safe and secure environment	4	6	-	-
PC6. identify and mitigate any safety and security hazards like illness, accidents, fires or acts of unlawful interference if it falls within the limits of the individuals authority	4	6	-	-
PC7. report any hazards outside the individuals authority to the relevant person in line with organisational procedures and regulatory guidelines	4	6	-	-
PC8. follow the organisations emergency procedures for accidents, fires or acts of unlawful interference	4	6	-	-
PC9. identify and recommend opportunities for improving health, safety, and security to the designated person	4	6	-	-
PC10. ensure that all health and safety records are updated and procedures are well defined	4	6	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1001
NOS Name	Follow organisation safety and security procedures
Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Machining
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	28/04/2022
Next Review Date	03/11/2027
NSQC Clearance Date	03/11/2022

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AAS/N1005: Carry-out pre-programming activities for aircraft parts

Description

This NOS unit describes the knowledge, understanding and skills required of an individual to carry out pre-programming activities as well as preparation for programming activities of special precision and complex geometry components

Scope

The scope covers the following :

- Pre-programming activities for machining components

Elements and Performance Criteria

Pre-programming activities for machining components

To be competent, the user/individual on the job must be able to:

- PC1.** read and understand the manufacturing drawing/ CAD model and its adaptability to CNC programming language
- PC2.** plan machining methodology for ease of manufacturing
- PC3.** plan the required tools, jigs and fixtures for supporting the component during machining
- PC4.** check the feasibility of machine for machining the components as per drawing and product output
- PC5.** develop the process and define the activities to be carried out on rawmaterial to obtain the finished output
- PC6.** document the discrepancies in model or drawing
- PC7.** select the type of CNC machine i.e 3-axis or 5-axis based on the criticality of the components to be machined
- PC8.** estimate the size of billet required to get the required output from machining
- PC9.** Check parameters, material characteristics and define machine axis speed, feed and cut to be taken during machining

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** different types of products manufactured by the company
- KU3.** different types of machining processes/ tools available
- KU4.** aerospace manufacturing standard and procedures such as ISO 9001, AS9100, etc
- KU5.** 3D CAD modelling
- KU6.** aerospace manufacturing standard and procedures
- KU7.** different types of machining processes and tolerances

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- KU8.** physical characteristics of various metals
- KU9.** basic fundamentals of machines and mechanics
- KU10.** different types of tools used in the machining process with respect to type of process to be conducted
- KU11.** basic principles of 5S in manufacturing
- KU12.** application of coolant and lubricants
- KU13.** impact of various machining processes on the final product outcome
- KU14.** basic arithmetic and calculation methods for tolerance limits
- KU15.** safety guidelines related to different machines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic level notes and observations
- GS2.** draw basic level drawing and charts
- GS3.** read and interpret work place related documentation
- GS4.** read and interpret engineering drawings and sketches
- GS5.** discuss task lists and job requirements with co-workers
- GS6.** effectively communicate information to team members
- GS7.** discuss with supervisor in order to understand the nature of the problem
- GS8.** attentively listen with full attention and comprehend the information given by the speaker
- GS9.** judge when to ask help from supervisor
- GS10.** suggest options to operators in case of any issue are observed during operation
- GS11.** plan work assigned on a daily basis and provides estimates of time required for each piece of work
- GS12.** ensure that customer needs are assessed and every effort is made to provide satisfactory service
- GS13.** recognise a work place problem or a potential problem and take action
- GS14.** determine problems needing priority action
- GS15.** gather information and provide assistance as required to solve problem
- GS16.** refer problem outside area of responsibility to approach person
- GS17.** analyse best possible solutions (cost, time, effort, etc.) suited for operations
- GS18.** analyse, evaluate and apply information gathered from observation, experience, reasoning or communication to act efficiently

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-programming activities for machining components</i>	45	55	-	-
PC1. read and understand the manufacturing drawing/ CAD model and its adaptability to CNC programming language	5	6	-	-
PC2. plan machining methodology for ease of manufacturing	5	6	-	-
PC3. plan the required tools, jigs and fixtures for supporting the component during machining	5	6	-	-
PC4. check the feasibility of machine for machining the components as per drawing and product output	5	6	-	-
PC5. develop the process and define the activities to be carried out on rawmaterial to obtain the finished output	5	6	-	-
PC6. document the discrepancies in model or drawing	5	6	-	-
PC7. select the type of CNC machine i.e 3-axis or 5-axis based on the criticality of the components to be machined	5	6	-	-
PC8. estimate the size of billet required to get the required output from machining	5	6	-	-
PC9. Check parameters, material characteristics and define machine axis speed, feed and cut to be taken during machining	5	7	-	-
NOS Total	45	55	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1005
NOS Name	Carry-out pre-programming activities for aircraft parts
Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Machining
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	15/05/2018
Next Review Date	03/11/2027
NSQC Clearance Date	03/11/2022

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AAS/N1006: Create CNC programs for aircraft parts

Description

This unit describes the procedure to create machining model and generate the program for different type of CNC controlled machine as per the complexity of machining

Scope

The scope covers the following :

- Creation of CNC program for machining the component

Elements and Performance Criteria

Creation of CNC program for machining the component

To be competent, the user/individual on the job must be able to:

- PC1.** evaluate 3D Model of the given aerospace component and related data
- PC2.** check the drawing/ model from machining perspective
- PC3.** define machining points, geometry, macros, machining parameters (speed, feed and depth of cut) and any check or island to be considered for not machining the specific area as indicated in the route card/ drawing/ model
- PC4.** generate the tool path and check with respect to model for digging or left out material as per colour code and change the program appropriately
- PC5.** create NC codes from apt source data by selecting appropriate post processor
- PC6.** carry out verification of the CNC program
- PC7.** transfer of NC codes to the shop floor for storage and input to the corresponding CNC machining centre
- PC8.** creation of NC layout sheet/ process sheet i.e. instruction sheet for the operator for NC machining.
- PC9.** hand over the controlled copy stamped layout sheet/process sheet to shop supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards & procedures followed in the company
- KU2.** different types of products manufactured by the company
- KU3.** different types of machining processes/ tools available
- KU4.** aerospace manufacturing standard and procedures such as ISO 9001, AS9100, etc
- KU5.** aerospace manufacturing standard and procedures
- KU6.** different types of machining processes
- KU7.** basic fundamentals of machines and mechanics

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- KU8.** different types of tools used in the machining process with respect to type of process to be conducted
- KU9.** basic principles of 5S in manufacturing
- KU10.** application of coolant and lubricants
- KU11.** impact of various machining processes on the final product outcome
- KU12.** basic arithmetic and calculation methods for tolerance limits
- KU13.** safety guidelines related to different machines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic level notes and observations
- GS2.** draw basic drawings and charts
- GS3.** read documents and notes
- GS4.** interpret/ comprehend the information given in the documents and notes
- GS5.** read and interpret symbols given on equipment and work area
- GS6.** discuss task lists and job requirements with co-workers
- GS7.** effectively communicate information to team members
- GS8.** question supervisor in order to understand the nature of the problem
- GS9.** attentively listen with full attention and comprehend the information given by the speaker
- GS10.** judge when to ask for help from a supervisor
- GS11.** suggest options to operators in case any issue is observed during operations
- GS12.** plan work assigned on a daily basis and provide estimates of time required for each piece of work
- GS13.** ensure that customer needs are assessed and every effort is made to provide satisfactory service
- GS14.** recognise a workplace problem or a potential problem and take action
- GS15.** determine problems needing priority action
- GS16.** refer problems outside area of responsibility to appropriate person
- GS17.** gather information and provide assistance as required to solve problems
- GS18.** analyse best possible solutions (cost, time, effort, etc.) suited for operations
- GS19.** analyse, evaluate and apply the information gathered from observation, experience, reasoning, or communicate to act efficiently

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Creation of CNC program for machining the component</i>	45	55	-	-
PC1. evaluate 3D Model of the given aerospace component and related data	5	7	-	-
PC2. check the drawing/ model from machining perspective	5	6	-	-
PC3. define machining points, geometry, macros, machining parameters (speed, feed and depth of cut) and any check or island to be considered for not machining the specific area as indicated in the route card/ drawing/ model	5	6	-	-
PC4. generate the tool path and check with respect to model for digging or left out material as per colour code and change the program appropriately	5	6	-	-
PC5. create NC codes from apt source data by selecting appropriate post processor	5	6	-	-
PC6. carry out verification of the CNC program	5	6	-	-
PC7. transfer of NC codes to the shop floor for storage and input to the corresponding CNC machining centre	5	6	-	-
PC8. creation of NC layout sheet/ process sheet i.e. instruction sheet for the operator for NC machining.	5	6	-	-
PC9. hand over the controlled copy stamped layout sheet/process sheet to shop supervisor	5	6	-	-
NOS Total	45	55	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1006
NOS Name	Create CNC programs for aircraft parts
Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Machining
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	15/05/2018
Next Review Date	03/11/2027
NSQC Clearance Date	03/11/2022

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AAS/N1007: Perform post-programming activities for aircraft parts

Description

This NOS unit describes the knowledge, understanding and skills required for an individual to perform post programming operations which include assuring correct billet, machine and clamping methods, correct tool and correct NC programme to be used during machining.

Scope

The scope covers the following :

- Check the correctness of inputs to the machine
- Assuring the quality of machined components

Elements and Performance Criteria

Check correctness of inputs to machine

To be competent, the user/individual on the job must be able to:

- PC1.** map the programmed machined component to the required output as per drawing/ model
- PC2.** check the required size of billet as per drawing/ model to minimise wastage
- PC3.** ensure suitable check points for inspection is provided in the programme
- PC4.** mention for use of devices like ultrasonic thickness measuring instrument, Vernier calipers, and other inspection equipment for measuring dimensions in stage machining
- PC5.** provide proper instructions to handle machined surface
- PC6.** ensure vacuum clamping , if required to be maintained and correct tool to be used by operator as per NC layout sheet (type of cutter, diameter, fillet radius, insert details etc.,)
- PC7.** discuss and take feedback from designers if any improvement/ correction of the program to be carried out as per inspection report

Assuring quality of machined components

To be competent, the user/individual on the job must be able to:

- PC8.** programme the correct datum of the component as per drawing / model
- PC9.** review the programme to ensure no digging or no wrong placement of clamps
- PC10.** take inputs on the programme from quality team to include stage inspection of the critical component
- PC11.** take the feedback if any improvement/ correction to the program to be carried out

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** different types of products manufactured by the company
- KU3.** different types of machining processes/ tools available

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- KU4.** aerospace manufacturing standard and procedures such as ISO 9001, AS9100, etc
- KU5.** aerospace manufacturing standard and procedures
- KU6.** different types of machining processes
- KU7.** basic fundamentals of machines and mechanics
- KU8.** different types of tools used in the machining process with respect to type of process to be conducted
- KU9.** basic principles of 5S in manufacturing
- KU10.** application of coolant and lubricants
- KU11.** Impact of various machining processes on the final product outcome
- KU12.** basic arithmetic and calculation methods for tolerance limits
- KU13.** safety guidelines related to different machines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic level notes and observations
- GS2.** draw basic level drawings and charts
- GS3.** read & comprehend documents and notes, process documentation & control panel
- GS4.** interpret/ comprehend the information given in the documents and notes
- GS5.** read and interpret symbols given on equipment and work area
- GS6.** read technology sheets/ HPS, sketches
- GS7.** discuss task lists and job requirements with co-workers
- GS8.** effectively communicate information to team members
- GS9.** question operator/ supervisor in order to understand the nature of the problem
- GS10.** attentively listen with full attention and comprehend the information given by the speaker
- GS11.** judge when to ask help from supervisor
- GS12.** suggest options to operators in case of any issue is observed during operation
- GS13.** plan work assigned on a daily basis and provide estimates of time required for each piece of work
- GS14.** ensure that customer needs are assessed and every effort is made to provide satisfactory service
- GS15.** recognise a work place problem or a potential problem and take action
- GS16.** determine problems needing priority action
- GS17.** gather information and provide assistance as required to solve problems
- GS18.** refer problems outside area of responsibility to approach person
- GS19.** analyse best possible solutions (cost, time, effort, etc.) suited for operations
- GS20.** analyse, evaluate and apply information gathered from observation, experience, reasoning or communication to act efficiently

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Check correctness of inputs to machine</i>	21	42	-	-
PC1. map the programmed machined component to the required output as per drawing/ model	3	6	-	-
PC2. check the required size of billet as per drawing/ model to minimise wastage	3	6	-	-
PC3. ensure suitable check points for inspection is provided in the programme	3	6	-	-
PC4. mention for use of devices like ultrasonic thickness measuring instrument, Vernier calipers, and other inspection equipment for measuring dimensions in stage machining	3	6	-	-
PC5. provide proper instructions to handle machined surface	3	6	-	-
PC6. ensure vacuum clamping , if required to be maintained and correct tool to be used by operator as per NC layout sheet (type of cutter, diameter, fillet radius, insert details etc.,)	3	6	-	-
PC7. discuss and take feedback from designers if any improvement/ correction of the program to be carried out as per inspection report	3	6	-	-
<i>Assuring quality of machined components</i>	13	24	-	-
PC8. programme the correct datum of the component as per drawing / model	3	6	-	-
PC9. review the programme to ensure no digging or no wrong placement of clamps	3	6	-	-
PC10. take inputs on the programme from quality team to include stage inspection of the critical component	3	6	-	-
PC11. take the feedback if any improvement/ correction to the program to be carried out	4	6	-	-
NOS Total	34	66	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1007
NOS Name	Perform post-programming activities for aircraft parts
Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Machining
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	15/05/2018
Next Review Date	03/11/2027
NSQC Clearance Date	03/11/2022

Qualification Pack

ASC/N0021: Maintain 5S at the work premises

Description

This NOS is about ensuring all 5 S activities both at the shop floor and the office area to facilitate increase in work productivity

Elements and Performance Criteria

Ensure sorting

To be competent, the user/individual on the job must be able to:

- PC1..** follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.
- PC2..** ensure segregation of waste in hazardous/ non hazardous waste as per the sorting work instructions
- PC3..** follow the technique of waste disposal and waste storage in the proper bins as per sop
- PC4..** segregate the items which are labelled as red tag items for the process area and keep them in the correct places
- PC5.** sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5s guidelines/ work instructions
- PC6. .** ensure that areas of material storage areas are not overflowing
- PC7.** properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required
- PC8.** return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area
- PC9.** follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards
- PC10.** follow the proper labeling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists

Ensure proper documentation and storage (organizing , streamlining)

To be competent, the user/individual on the job must be able to:

- PC11.** check that the items in the respective areas have been identified as broken or damaged
- PC12.** follow the given instructions and check for labelling of fluids, oils. lubricants, solvents, chemicals etc. and proper storage of the same to avoid spillage, leakage, fire etc
- PC13.** make sure that all material and tools are stored in the designated places and in the manner indicated in the 5s instructions

Ensure cleaning of self and the work place

To be competent, the user/individual on the job must be able to:

- PC14.** check whether safety glasses are clean and in good condition
- PC15.** keep all outside surfaces of recycling containers are clean
- PC16..** ensure that the area has floors swept, machinery clean and generally clean. in case of cleaning, ensure that proper displays are maintained on the floor which indicate potential safety hazards

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- PC17..** check whether all hoses, cabling & wires are clean, in good condition and clamped to avoid any mishap or mix up
- PC18..** ensure workbenches and work surfaces are clean and in good condition
- PC19.** follow the cleaning schedule for the lighting system to ensure proper illumination
- PC20.** store the cleaning material and equipment in the correct location and in good condition
- PC21.** ensure self-cleanliness - clean uniform, clean shoes, clean gloves, clean helmets, personal hygiene

Ensure sustenance

To be competent, the user/individual on the job must be able to:

- PC22.** follow the daily cleaning standards and schedules to create a clean working environment
- PC23.** attend all training programs for employees on 5 s
- PC24.** support the team during the audit of 5 s
- PC25.** participate actively in employee work groups on 5s and encourage team members for active participation
- PC26.** follow the guidelines for what to do and what not to do to build sustainability in 5s as mentioned in the 5s check lists/ work instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards, procedures and policies related to 5S followed in the company
- KU2.** have basic knowledge of 5S procedures
- KU3.** know various types 5s practices followed in various areas
- KU4.** understand the 5S checklists provided in the department/ team
- KU5.** have skills to identify useful & non useful items
- KU6.** have knowledge of labels , signs & colours used as indicators
- KU7.** knowledge on how to sort and store various types of tools, equipment, material etc.
- KU8.** know , how to identify various types of waste products
- KU9.** understand the impact of waste/ dirt/ dust/unwanted substances on the process/ environment/ machinery/ human body
- KU10.** have knowledge of best ways of cleaning & waste disposal
- KU11.** understand the importance of standardization in processes
- KU12.** understand the importance of sustainability in 5S
- KU13.** have knowledge of TQM process
- KU14.** have knowledge of various materials and storage norms
- KU15.** understand visual controls, symbols, graphs etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic level notes and observations

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- GS2.** note down observations (if any) related to the process
- GS3.** read 5S instructions put up across the plant premises
- GS4.** effectively communicate information to team members inform employees in the plant and concerned functions about 5S
- GS5.** question the process head in order to understand the 5S related issues
- GS6.** attentively listen with full attention and comprehend the information given by the speaker during 5S training programs
- GS7.** use common sense and make judgments during day to day basis
- GS8.** use reasoning skills to identify and resolve basic problems using 5S
- GS9.** persuade co team members to follow 5 S
- GS10.** ensure that the co team members understand the importance of using 5 S tool
- GS11.** use innovative skills to perform and manage 5 S activities at the work desk and the shop floor
- GS12.** exhibit inquisitive behaviour to seek feedback and question on the existing set patterns of work
- GS13.** do what is right, not what is a popular practices
- GS14.** follow shop floor rules& regulations and avoid deviations; make 5S an integral way of life
- GS15.** ensure self-cleanliness on a daily basis
- GS16.** demonstrate the will to keep the work area in a clean and orderly manner

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure sorting</i>	10	30	-	-
PC1.. follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.	1	3	-	-
PC2.. ensure segregation of waste in hazardous/ non hazardous waste as per the sorting work instructions	1	3	-	-
PC3.. follow the technique of waste disposal and waste storage in the proper bins as per sop	1	3	-	-
PC4.. segregate the items which are labelled as red tag items for the process area and keep them in the correct places	1	3	-	-
PC5. sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5s guidelines/ work instructions	1	3	-	-
PC6. . ensure that areas of material storage areas are not overflowing	1	3	-	-
PC7. properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required	1	3	-	-
PC8. return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area	1	3	-	-
PC9. follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards	1	3	-	-
PC10. follow the proper labeling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure proper documentation and storage (organizing , streamlining)</i>	3	9	-	-
PC11. check that the items in the respective areas have been identified as broken or damaged	1	3	-	-
PC12. follow the given instructions and check for labelling of fluids, oils. lubricants, solvents, chemicals etc. and proper storage of the same to avoid spillage, leakage, fire etc	1	3	-	-
PC13. make sure that all material and tools are stored in the designated places and in the manner indicated in the 5s instructions	1	3	-	-
<i>Ensure cleaning of self and the work place</i>	8	24	-	-
PC14. check whether safety glasses are clean and in good condition	1	3	-	-
PC15. keep all outside surfaces of recycling containers are clean	1	3	-	-
PC16.. ensure that the area has floors swept, machinery clean and generally clean. in case of cleaning, ensure that proper displays are maintained on the floor which indicate potential safety hazards	1	3	-	-
PC17.. check whether all hoses, cabling & wires are clean, in goodcondition and clamped to avoid any mishap or mix up	1	3	-	-
PC18.. ensure workbenches and work surfaces are clean and in good condition	1	3	-	-
PC19. follow the cleaning schedule for the lighting system to ensure proper illumination	1	3	-	-
PC20. store the cleaning material and equipment in the correct location and in good condition	1	3	-	-
PC21. ensure self-cleanliness - clean uniform, clean shoes, clean gloves, clean helmets, personal hygiene	1	3	-	-
<i>Ensure sustenance</i>	4	12	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. follow the daily cleaning standards and schedules to create a clean working environment	1	3	-	-
PC23. attend all training programs for employees on 5 s	0.5	2	-	-
PC24. support the team during the audit of 5 s	1	3	-	-
PC25. participate actively in employee work groups on 5s and encourage team members for active participation	0.5	2	-	-
PC26. follow the guidelines for what to do and what not to do to build sustainability in 5s as mentioned in the 5s check lists/ work instructions	1	2	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N0021
NOS Name	Maintain 5S at the work premises
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

ASC/N9801: Organize work and resources (Service)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising use of resources

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Perform work as per quality standards
- Health and hygiene
- Material/energy conservation practices
- Effective waste management practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** organise work as per organisation's current health, safety and security policies and procedures
- PC2.** report any identified breaches in health, safety, and security policies and procedures to the designated person
- PC3.** identify the risks and hazards associated with work activities, their causes and prevention

Perform work as per quality standards

To be competent, the user/individual on the job must be able to:

- PC4.** ensure work area is clean and tidy
- PC5.** ensure that work is accomplished as per the requirements within the specified timeline
- PC6.** ensure team goals are given preference over individual goals

Health and hygiene

To be competent, the user/individual on the job must be able to:

- PC7.** sanitize workstation and equipment regularly
- PC8.** clean hands with soap, alcohol-based sanitizer regularly
- PC9.** avoid contact with ill people and self-isolate in a similar situation
- PC10.** wear and dispose PPEs regularly and appropriately
- PC11.** report advanced hygiene and sanitation issues to appropriate authority
- PC12.** follow stress and anxiety management techniques

Material/energy conservation practices

To be competent, the user/individual on the job must be able to:

- PC13.** identify ways to optimise usage of material in various tasks/activities/processes
- PC14.** use resources, including water, in a responsible manner

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- PC15.** check for spills/leakages in various tasks/activities/processes
- PC16.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC17.** carry out routine cleaning of tools, machines and equipment
- PC18.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC19.** report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment
- PC20.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Effective waste management practices

To be competent, the user/individual on the job must be able to:

- PC21.** identify recyclable and non-recyclable, and hazardous waste generated
- PC22.** segregate waste into different categories
- PC23.** dispose non-recyclable waste appropriately
- PC24.** deposit recyclable and reusable material at identified location
- PC25.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, and individual role and responsibilities in this context
- KU2.** the organisations emergency procedures for different emergency situations and the importance of following the same
- KU3.** evacuation procedures for workers and visitors
- KU4.** how and when to report hazards as well as the limits of responsibility for dealing with hazards
- KU5.** potential hazards, risks and threats based on the nature of work
- KU6.** the implications of own work on the schedule and work of others
- KU7.** efficient utilisation of material and water
- KU8.** basics of electricity and prevalent energy efficient devices
- KU9.** ways to recognise common electrical problems
- KU10.** common practices of conserving electricity
- KU11.** common sources of pollution and ways to minimize it
- KU12.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU13.** usage of different colours of dustbins
- KU14.** waste management and methods of waste disposal
- KU15.** significance of greening
- KU16.** organisation's policies to maintain personal health and hygiene at workplace

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/standard operating procedures
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** modify work practices to improve them
- GS4.** ask for clarifications from superior about the job requirement
- GS5.** work with supervisors/team members to carry out work related tasks
- GS6.** complete tasks efficiently and accurately within stipulated time
- GS7.** inform/report to concerned person in case of any problem
- GS8.** make timely decisions for efficient utilization of resources
- GS9.** write in at least one language and complete written work with attention to detail
- GS10.** record data on waste disposal at workplace
- GS11.** be punctual, utilize time and manage workload efficiently
- GS12.** evaluate strategies to maintain, enhance or reduce the intensity of heightened emotional response

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	8	4	-	3
PC1. organise work as per organisation's current health, safety and security policies and procedures	-	2	-	1
PC2. report any identified breaches in health, safety, and security policies and procedures to the designated person	3	1	-	-
PC3. identify the risks and hazards associated with work activities, their causes and prevention	5	1	-	2
<i>Perform work as per quality standards</i>	12	8	-	6
PC4. ensure work area is clean and tidy	4	2	-	-
PC5. ensure that work is accomplished as per the requirements within the specified timeline	6	4	-	2
PC6. ensure team goals are given preference over individual goals	2	2	-	4
<i>Health and hygiene</i>	12	8	-	5
PC7. sanitize workstation and equipment regularly	2	2	-	2
PC8. clean hands with soap, alcohol-based sanitizer regularly	2	1	-	-
PC9. avoid contact with ill people and self-isolate in a similar situation	2	1	-	-
PC10. wear and dispose PPEs regularly and appropriately	2	2	-	1
PC11. report advanced hygiene and sanitation issues to appropriate authority	2	2	-	2
PC12. follow stress and anxiety management techniques	2	-	-	-
<i>Material/energy conservation practices</i>	10	4	-	3
PC13. identify ways to optimise usage of material in various tasks/activities/processes	2	-	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. use resources, including water, in a responsible manner	2	-	-	-
PC15. check for spills/leakages in various tasks/activities/processes	-	1	-	-
PC16. plug spills/leakages and escalate to appropriate authority if unable to rectify	-	1	-	1
PC17. carry out routine cleaning of tools, machines and equipment	2	-	-	-
PC18. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	-	1	-	1
PC19. report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment	2	1	-	-
PC20. ensure electrical equipment and appliances are properly connected and turned off when not in use	2	-	-	-
<i>Effective waste management practices</i>	8	6	-	3
PC21. identify recyclable and non-recyclable, and hazardous waste generated	2	-	-	1
PC22. segregate waste into different categories	-	2	-	-
PC23. dispose non-recyclable waste appropriately	2	2	-	1
PC24. deposit recyclable and reusable material at identified location	2	1	-	-
PC25. follow processes specified for disposal of hazardous waste	2	1	-	1
NOS Total	50	30	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9801
NOS Name	Organize work and resources (Service)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

ASC/N9802: Interact effectively with colleagues, customers and others

Description

This NOS unit is about communicating with customers and colleagues/superiors, either in own work group or in other work groups within organisation.

Scope

The scope covers the following :

- Communicate effectively with colleagues, customers and others
- Interact with supervisor or superior

Elements and Performance Criteria

Communicate effectively with colleagues, customers and others

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues, customers and others, wherever needed, through all means i.e. face-to-face, telephonic or written
- PC2.** adjust communication styles to reflect gender and persons with disability (PwD) sensitivity
- PC3.** work in a way that shows respect for colleagues and others
- PC4.** follow the organisation's policies and procedures while working in a team
- PC5.** respect personal space of colleagues and customers

Interact with supervisor or superior

To be competent, the user/individual on the job must be able to:

- PC6.** identify work requirements by receiving instructions from reporting supervisor
- PC7.** escalate problems to supervisors that cannot be handled including repairs and maintenance of machine
- PC8.** report the completed work
- PC9.** rectify errors as per feedback

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with colleagues and supervisor
- KU2.** different methods of communication as per the circumstances
- KU3.** gender based concepts, issues and legislation

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues and customers
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** deliver consistent and reliable service to customers
- GS7.** complete written work with attention to detail
- GS8.** check that the work meets customer requirements

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with colleagues, customers and others</i>	36	11	-	14
PC1. maintain clear communication with colleagues, customers and others, wherever needed, through all means i.e. face-to-face, telephonic or written	8	-	-	4
PC2. adjust communication styles to reflect gender and persons with disability (PwD) sensitivity	8	-	-	-
PC3. work in a way that shows respect for colleagues and others	7	4	-	3
PC4. follow the organisation's policies and procedures while working in a team	7	4	-	3
PC5. respect personal space of colleagues and customers	6	3	-	4
<i>Interact with supervisor or superior</i>	14	19	-	6
PC6. identify work requirements by receiving instructions from reporting supervisor	7	4	-	-
PC7. escalate problems to supervisors that cannot be handled including repairs and maintenance of machine	-	5	-	3
PC8. report the completed work	7	5	-	-
PC9. rectify errors as per feedback	-	5	-	3
NOS Total	50	30	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9802
NOS Name	Interact effectively with colleagues, customers and others
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
AAS/N0503.Work Effectively in a Team	45	55	0	0	100	5
AAS/N1001.Follow organisation safety and security procedures	40	60	-	-	100	10
AAS/N1005.Carry-out pre-programming activities for aircraft parts	45	55	0	0	100	20
AAS/N1006.Create CNC programs for aircraft parts	45	55	0	0	100	25
AAS/N1007.Perform post-programming activities for aircraft parts	34	66	0	0	100	20
ASC/N0021.Maintain 5S at the work premises	25	75	-	-	100	10
ASC/N9801.Organize work and resources (Service)	50	30	-	20	100	5
ASC/N9802.Interact effectively with colleagues, customers and others	50	30	-	20	100	5
Total	334	426	-	40	800	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.